SACHIN GAYAKWAD

LEAD ENGINEER

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Mechanical Lead Engineer and Strategic Engineering Leader with 12+ years of experience in electro-mechanical design, structural engineering, product lifecycle management (PLM), and global NPI delivery. Expert in system integration, Project management, Modify Launched Design (MLD), sustaining engineering, DFMEA, DFX, GD&T, CTS/CTQ, tolerance stack-up, and design controls aligned with UL/IEC/CE regulatory standards. Certified Six Sigma Black Belt and PMP with strong capability in feasibility studies, risk management, value engineering (VE), complexity reduction, and supplier collaboration. Proven track record in driving architecture definition, cross-functional leadership, cost optimization, manufacturability improvements, and data-driven problem solving to deliver reliable, scalable, and compliant product platforms across global electro-mechanical programs.

EDUCATION

Master's in Design Eng.: Design Engineering

BITS PILANI, Pune, 2023, GPA: 8.7

B.E. Mechanical Engineering, Visvesvaraya Technological University, Belgaum, 2013, Percentile: 82%

CORE COMPETENCIES

- Mechanical System Design & Integration
- Product Development & Lifecycle Management (PLM)
- Continuous Improvement, Six Sigma
- Supplier Relationship Management
- Risk Assessment and Mitigation
- Innovation and Design Thinking
- Process Map, Kaizen & Six Sigma
- PFMEA, DFMEA, DFX, DFM, Creo, Minitab, Cetol
- System Architecture & Requirement management
- Design verification & Validation
- People leadership, Hiring & Performance review
- Strategic Technology Roadmap, Complexity management & ZBP

PROFILE SUMMARY

- Accomplished **Mechanical Engineering Leader** with nearly **12 years of progressive experience** in the design, development & integration of mechanical systems for home appliances & switchgear products.
- **Currently working as Lead Engineer at Whirlpool Corporation**, overseeing the integration of complex **mechanical systems** and leading cross-functional teams to ensure successful **product launches**, operational efficiency, continuous improvement across global programs.
- Experienced in Structural Design with strong expertise in **plastics, sheet metal,** and **casting** components, backed by solid material knowledge. Proficient in **DFMEA, DFX, PPAP,** and **APQP** to ensure high reliability, manufacturability, and compliance with **UL, CB & IEC.**
- Proficient in CI and systematic problem-solving tools, including **Six Sigma, DOE, RCA**, and thermal characterization to solve complex **structural**, **mechanical,** and **thermal** problems in cooking appliances and enhance product performance
- Demonstrated capability in driving **project governance** through accurate s**cheduling**, risk assessment, requirement **scoping**, IDP development, and **business case analysis**, enabling transparent decision-making and effective **stakeholder management.**

ACCOMPLISHMENTS

- Accelerated product delivery timelines by **22%** through continuous improvement initiatives in finished product sourcing, enabling time-to-market in **under six months.**
- Achieved **$12M** in savings by leveraging **VAVE**, **feasibility analysis**, and structured **cost-ideation strategies** to drive structural design optimization for **cooking products.**
- Created **WBS, project charters, resource plans, and baseline schedules** to guide execution teams for more than 10 projects.
- Enhanced **hood production capacity by 1.5×** by applying lean and waste-reduction techniques.
- Delivered **5 high-impact projects** and **10+ new product** introductions, while managing multiple maintenance and sustaining engineering programs across **NAR, EMEA**, and **Asia** regions.
- Contributed to research and innovation initiatives resulting in **two patents granted for the NAR (North America Region)**, showcasing technical excellence and thought leadership.
- Drove **micro-switch** endurance improvement to **500K** cycles using structured design optimization and root-cause elimination, removing the need **for higher-life variants**.

CAREER TIMELINE

2013-14 — IEW CRANES PVT. LTD.

2014-16 — ELCOM INTERNATIONAL

2017-Present — Whirlpool CORPORATION

CERTIFICATIONS

- Project Management Professional (PMP)
- Operational Excellence (OPEX) Black Belt
- Six Sigma Green Belt
- Design for Safety – Level 4
- Reliability Engineering

SOFT SKILLS

- Structured Problem-solving
- Decision-making
- Cross Functional Leadership
- Team Management, Technical Mentorship & Coaching
- Adaptability & Change management
- Stakeholder Management & Communication

PERSONAL DETAILS

Address: Pune, India

Date of Birth: 30th July, 1990

Language Known: English, Hindi, Marathi & Kannada

Passport Details: Passport number: ZA153097

Passport validity: 22nd May,2035

Nationality: Indian

Driving Details: LMV/ MCWG

Validity: 10th July, 2037

Marital Status: Married

Number of dependencies: 3

Visa Status: USA B1/B2 Active till Sep' 2027

WORK EXPERIENCE

Lead Engineer – Mechanical System Integrator | Whirlpool Corporation | Pune | Jan 2017 – Present

Key Roles and Responsibilities:

- Spearheading **end-to-end design, integration, and system validation** of complex home appliance systems — including **hoods, cooktops, Built-in ovens, FSRs, warming drawers, washing machines, and MWOs** — across multiple global programs.
- Managed completely new product development lifecycle, covering **requirement** analysis, target definition, **concept engineering, subsystem integration**, and **validation** testing, ensuring **reliable** and **high-quality** product launches.
- Functioning as **Technical Lead** overseeing multidisciplinary teams, ensuring seamless **mechanical–electrical–software integration** and product delivery aligned with strategic goals.
- Developing and executing **technical** and **capability roadmaps** to drive innovation, technology maturity, complexity reduction, Zero Base Planning **(ZBP)**, and **value-stream optimization** across **mechanical systems and subassemblies**.
- Ensuring **global compliance with regulatory standards,** including **UL, COC, CB, and IEC**, enabling product certification across the **Asia and North America regions.**
- Delivering **data-driven trade-off analyses** balancing design, cost, quality, time constraints; ensured adherence to product plan guardrails through agile technical decision-making.
- Playing a key role in **requirements analysis, scoping, effort estimation, and project execution**, collaborating closely with global cross-functional and supplier teams.
- Defining the **mechanical capability roadmap** and **establishing lab infrastructure** for thermal, regulatory, and structural testing to enable comprehensive in-house validation.
- **Performing mechanical hand calculations** for stiffness, thermal, and load analyses to validate design robustness prior to FEA.
- Leading **recruitment**, **onboarding**, and **capability development initiatives**, while driving coaching, technical mentoring, and succession planning to build a **high-performing engineering talent pipeline**.
- **Directed strategic product launches** through finished product sourcing initiatives, achieving an **$8M DCM impact**, driving cost optimization and operational efficiency.
- **Mentored 10+ engineers** in delivering subsystem-level designs, interface management, and performance optimization, cultivating a high-performance engineering culture.

Senior Design & Development Engineer | Elcom International | Kolhapur | Jun 2014 – Dec 2016

- Delivered **innovative engineering solutions** leveraging advanced problem-solving methodologies such as **MSE, COV, DOE, and Noise Strategy** to optimise design robustness and performance.
- Drove **line complexity reduction initiatives**, achieving a **28% improvement in production cycle time** for micro switch assembly processes.
- Translated **2D design concepts into 3D solid models**, ensuring dimensional accuracy, design intent, and compliance with client specifications.
- Proficient in **design simulation, tolerance stack-up and statistical validation** using **Cetol, Creo, Minitab, SAP, SolidWorks, Lucidchart, DFMPro** for precision-driven design validation.
- Drove **line complexity reduction initiatives**, achieving a **28% improvement in production cycle time** for micro switch assembly processes.
- Translated **2D design concepts into 3D solid models**, ensuring dimensional accuracy.

Design Engineer | IEW Cranes, Pune | Jun 2013 – May 2014

- Designed and developed **5T Hoists, Goliath Cranes, and Jib Cranes**, optimizing performance, safety, and manufacturing feasibility.
- Applied **Pro/E–Creo** for 3D modeling, **BOM generation & cost analysis**, ensuring accuracy.
- Collaborated closely with the Manufacturing Team to ensure **design-to-production transition**, reducing lead-time and enhancing product quality.
- Gained extensive exposure to **plastic and sheet metal component design**, strengthening multi-domain design versatility.